

Local Search with Correlated Randomness

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Abstract

How much does a randomized algorithm’s running-time distribution reveal about its behavior under a different source of randomness? For retry tasks, every selection rule has the same stopping law under independent randomness, and the same prefix power sums at every depth and power. Yet two orders have worst-case failure probabilities 1 and $\exp[-\Theta(n)]$ at the same linear deadline. We identify an optimal priority rule for arbitrary heterogeneous tasks and every deadline.

We consider correlated sources satisfying $\nu[w] \leq DP[w]^s$ for every execution prefix w , relative to an independent reference law P . This cumulative condition allows long predictable stretches. Classical tree max-flow/min-cut expresses the worst-case probability of a failed run in terms of the minimum cost of a cover by prefixes of different lengths. We develop the algorithmic consequences of retaining this prefix structure.

For the standard walk on a k -CNF with an assignment making at least r literals true in every clause, where $k/2 < r < k$, we determine the sharp prefix-entropy threshold. At the threshold, the expected number of flips is $O_{k,r}((h + \log D + 1)^3)$ from distance h to a strongly satisfying assignment. This dependence is optimal in both h and D and holds under arbitrary clause overlap and history-dependent selection. Given a distance bound $H \geq h$, an observable Hamming-ball restart rule gives an exponential tail on scale H^2 , without knowing the witness or receiving fresh randomness. More generally, we characterize exactly when restarts restore exponential tails under a single globally constrained stream and construct a stationary rule on explicit finite state spaces.

1 Introduction

How much does a randomized algorithm’s running-time distribution reveal about its behavior under a different source of randomness? The distribution tells us how much probability survives at each time. It does not tell us which earlier choices lead to those surviving runs. We show that this missing information can determine both the robustness of an algorithm and how it should be designed.

Consider several tasks that can each be retried until they succeed. Under independent randomness, every rule for choosing the next unfinished task has exactly the same total running-time distribution. The agreement goes further: at every depth, the sums of all positive powers of the surviving prefix probabilities are identical. Yet under the same class of correlated sources, one order can fail with probability one at a linear deadline while another fails with probability only $\exp[-\Theta(n)]$. We identify an optimal order, simultaneously for every deadline, by examining how the execution prefixes fit together. Thus information discarded by all of these single-depth summaries can change which algorithm one should run.

A global guarantee on randomness. Our source assumption controls the probability of every complete prefix, rather than the next choice conditional on the past. The distinction is already visible in a simple bit stream: take m independent fair bits, then m deterministic zeroes, then independent fair bits forever. Every length- t prefix has probability at most $2^{-t/2}$. Nevertheless, the entire middle block is predictable. An algorithm that reaches this block has no fresh randomness to use, and restarting it does not make the next bit fair.

More generally, fix the primitive random choices of an algorithm and their independent reference law P . We consider all laws ν on the complete choice stream satisfying

$$\nu[w] \leq DP[w]^s \quad \text{for every finite prefix } w,$$

where $0 < s \leq 1$ and $D \geq 1$. For uniform choices, this is a lower bound on the min-entropy of each initial block. It allows dependence within and across blocks, including the deterministic stretch above. The question is how well a specified algorithm works with this cumulative guarantee. Its random choices and their interpretation are kept fixed.

This formulation separates two issues that an independence assumption usually resolves at once: how concentrated the choice stream can be, and what randomness remains after an observed execution history. Conditioning can weaken the remaining guarantee by an arbitrarily large factor. Consequently, an argument based on conditional drift or on independent restart attempts cannot simply be reused. A proof must instead account for the complete histories on which the algorithm fails. Questions about the quality and quantity of randomness in SAT local search have also been studied experimentally [TH06]; here we ask for worst-case guarantees under an explicit source condition. A finite-deadline guarantee needs this condition only through that deadline.

What a prefix cover records. A failed run may become expensive to produce well before the deadline. Different runs may do so at different times. A prefix cover selects these earlier histories and sums their powered reference probabilities. Unlike a count taken at a single depth, it retains how surviving histories extend one another. Classical tree max-flow/min-cut translates the minimum cover cost into the exact worst-case failure probability allowed by the source condition. We use this identity to analyze a familiar SAT walk, characterize the value of restarting, and prove the optimal retry priority described above.

The exact cost of correlation for a SAT walk. Suppose an unknown assignment x^* makes at least r literals true in every width- k clause, where $k/2 < r < k$. The standard walk selects a violated clause and flips a uniformly indexed variable in it. Clauses may overlap arbitrarily, and the selector may use the complete history. Under independent randomness this is an established algorithm [AGH17, Proposition 6.1], requiring $O_{k,r}(h)$ expected flips from distance h to x^* . At the sharp global prefix-entropy threshold, we prove the bound

$$\mathbb{E}_\nu \tau = O_{k,r}((h + \log D + 1)^3).$$

The dependence on both h and D is optimal. On the matching family, one admissible source forces a delay of order h^3 with probability one for large h , for every clause selector. The algorithm knows neither x^* nor h , and the upper bound is independent of the number of variables.

The comparison is sharper than polynomial termination alone. A single-depth spectral argument already gives $O_{k,r}(n^2(h + \log D + 1))$ expected flips. The new bound removes that ambient-size factor. It also gives a stretched-exponential tail on a family where every single-depth cover costs at least a polynomial amount. The upper proof uses an integer progress measure on the full execution tree. The matching lower bound follows by tracking a region of the capacity recurrence whose value remains one for a cubic time.

Turning the bound into a better algorithm. The analysis identifies long excursions as the obstacle to an exponential tail. Suppose a bound $H \geq h$ is available. Run the same SAT walk, but reset to its initial assignment whenever an unsatisfying assignment leaves the Hamming ball of radius $2H$ around it. This rule uses neither x^* nor an enumeration of the state space. It preserves the progress comparison while confining the unknown witness distance to at most $3H$.

The resulting failure probability is at most $\min\{1, CD\lambda^h e^{-cT/H^2}\}$, where $\lambda = \sqrt{r/(k-r)}$ and c, C depend only on k, r . The guarantee holds for arbitrary clause overlap and history-dependent selection. The source continues across resets; the proof never assumes independent attempts.

An exact criterion for restarting. The general principle is simple. A finite prefix cover of unsuccessful runs with cost $c < 1$ specifies when to restart. Concatenating j such attempts gives total powered cost at most c^j , and hence an exponential tail under one global source constraint. We prove a converse: some restart rule ensures almost-sure termination for every admissible unit-constant source if and only if the original nontermination event has cover capacity below one. The same condition ensures an exponential tail for every finite D . A linear program constructs a stationary rule on an explicit finite state space. A three-state example separates an adaptive rule from every predetermined sequence of cutoff times.

Sharing a random stream. Sharing a source requires its own analysis. There are components with exponentially small individual worst-case nontermination probabilities whose combined execution is made nonterminating by one admissible source under every interleaving. The source condition is global; it need not hold on each component's subsequence. The ordering and composition results both depend on the position of choices within the full execution.

Sections 2–6 give the model and main results. Appendices A–E contain their complete proofs, organized by topic. ¹

¹The AI methodologies statement at the end describes the assistance used in this work.

2 Prefix covers and source robustness

Fix a finite alphabet $\mathcal{A}$ with at least two symbols and a full-support distribution π . Let $P = \pi^{\mathbb{N}}$. A random tape is a sequence in $\mathcal{A}^{\mathbb{N}}$; $[w]$ is the set of tapes beginning with a finite word w , and $P[w]$ is its reference probability. The primitive alphabet and its interpretation by the algorithm are fixed. Splitting or recoding symbols changes the source assumption.

For $0 < s \leq 1$ and $D \geq 1$, define

$$\mathcal{Q}_{s,D} = \{\nu : \nu[w] \leq DP[w]^s \text{ for every finite } w\}. \quad (1)$$

Here ν ranges over probability laws on whole tapes, and one constant D applies at all lengths. For the uniform law on q symbols, this is exactly the requirement that the first t symbols have min-entropy at least $st \log_2 q - \log_2 D$. A conditional bound q^{-s} on the probability of each next symbol implies membership in $\mathcal{Q}_{s,1}$; the converse is false, as the fair-bit example in the introduction shows. Under a nonuniform reference, (1) is a relative probability bound rather than a fixed-rate entropy bound. Conditioning on a positive-probability prefix w gives $\nu[v \mid w] \leq D_w P[v]^s$, where $D_w = DP[w]^s / \nu[w]$. This constant can be much larger than D ; there is no uniform reset of the original guarantee after a history.

A selector is deterministic and *nonanticipating*: it uses the observed history but no future symbols. Let τ count updates before termination. An update attempts to resolve a violated constraint; it need not decrease the total number of violations. Write $\mathcal{N} = \{\tau = \infty\}$. For a closed tape event E , define

$$\mathcal{C}_s(E) = \inf_{\mathcal{B}} \sum_{w \in \mathcal{B}} P[w]^s, \quad E \subseteq \bigcup_{w \in \mathcal{B}} [w], \quad (2)$$

where $\mathcal{B}$ ranges over finite prefix-free covers. Prefix-free means that no chosen word extends another, so the cylinders are disjoint. Their lengths may differ. The empty word has cost one.

Theorem 2.1 (Classical tree capacity identity). *For every closed E , $0 < s \leq 1$, and $1 \leq D < \infty$,*

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu(E) = \min\{1, D\mathcal{C}_s(E)\}. \quad (3)$$

If E depends on the first T symbols, a minimizing cover uses words of length at most T . For finite-symbol events $E_T \downarrow \mathcal{N}$,

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu(\mathcal{N}) = \lim_{T \rightarrow \infty} \max_{\nu \in \mathcal{Q}_{s,D}} \nu(E_T). \quad (4)$$

A source is a probability flow down the prefix tree; a cover cuts every branch in E . Tree max-flow/min-cut gives the identity, as in the tree proof of Frostman's lemma [BP17, Section 3.1]. Appendix A includes the completion of a maximizing subflow to a probability law.

What a single-depth bound retains. Let $\mathcal{L}_T$ be the minimal prefixes completing T updates, including a possible final update that terminates the algorithm. Put

$$Z_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s. \quad (5)$$

Then $Z_T(1) = P\{\tau \geq T\}$ and $\mathbb{P}_\nu\{\tau \geq T\} \leq DZ_T(s)$. With one symbol per update, let $\tilde{Z}_T(s)$ instead sum over the length- T words still live after the update. For a uniform q -symbol source,

$$Z_T(s) = q^{(1-s)T} P\{\tau \geq T\}. \quad (6)$$

These sums count each level separately. A prefix cover can combine levels and therefore uses how the surviving prefixes extend one another.

The asymptotic geometric threshold. The metric $d_P(\omega, \eta) = P[\omega \wedge \eta]$ for distinct tapes gives cylinder $[w]$ diameter $P[w]$. The Hausdorff dimension $\delta = \dim_H^P \mathcal{N}$ is the critical covering exponent [Bil65, Lut11]. For $s < \delta$, some finite D allows a source in $\mathcal{Q}_{s,D}$ supported on $\mathcal{N}$; for $s > \delta$, every such source assigns it probability zero. At equality, and at finite deadlines, the cover cost is needed as well as dimension. Corollary A.1 gives this connection precisely.

3 Local SAT search: sharp bounds and restarts

A local-search flip can satisfy one clause while violating several others. We now analyze this interaction for the standard promise-SAT random walk. Unlike the examples with an exact one-dimensional state reduction, the bound applies to arbitrary overlapping formulas. Its cost depends on the distance from the initial assignment to a solution, even when the source correlates its choices across the entire execution.

Fix integers $k/2 < r < k$. A k -CNF is *strongly r -satisfiable* if some assignment x^* makes at least r literals true in each clause; clauses use k distinct variables. The algorithm chooses any violated clause and flips the variable indexed by the next source symbol. Under the reference source this index is uniform. The clause choice may depend on the complete past. This is an established polynomial-time algorithm for the promise problem [AGH17, Proposition 6.1]. We fix an ordering of each clause's literals and take source symbols in $[k]$, with uniform reference law P . Put

$$\lambda = \sqrt{\frac{r}{k-r}} > 1, \quad s_* = \log_k(2\sqrt{r(k-r)}) \in (0, 1). \quad (7)$$

Theorem 3.1 (Robustness from distance to a solution). *Let Φ be a strongly r -satisfiable k -CNF on n variables, and let h be the Hamming distance of its initial assignment from a strongly satisfying assignment x^* . There are constants $c, C > 0$ depending only on k, r such that, for every deterministic nonanticipating clause selector, every $D \geq 1$, every $\nu \in \mathcal{Q}_{s_*, D}$, and every integer $T \geq 0$,*

$$\mathbb{P}_\nu\{\tau > T\} \leq \min\{1, CD\lambda^h e^{-cT^{1/3}}\}, \quad \mathbb{E}_\nu \tau \leq C(h + \log D + 1)^3. \quad (8)$$

Here τ counts variable flips until the formula is satisfied. There is no bound on the number or overlap degree of clauses. The same conclusions hold at every $s \in [s_*, 1]$.

The walk does not know x^* . Under the ideal source its expected number of flips is at most $kh/(2r-k)$. Theorem 3.1 bounds the loss when independence is replaced by a global prefix constraint. In particular, the robust bound remains small when an initial assignment is close to a strong solution, regardless of how many other variables the formula contains. The statement counts flips; it does not suppress the work required to find a violated clause.

Why different depths help. If a selected clause is violated, at least r of its variables disagree with x^* . Thus at least r labels decrease the distance and the others increase it. The distance process need not be Markov: the number of decreasing labels can vary with the clause and the entire history. At s_* the comparison weights are $a = rk^{-s_*} = \lambda/2$ and $b = (k-r)k^{-s_*} = 1/(2\lambda)$. For an integer $m > h$, use prefixes that first reach distance m and length- T prefixes that remain below it. The first set has powered mass at most $(h/m)\lambda^{h-m}$; a reflected spectral comparison bounds the second by $C\lambda^h e^{-cT/m^2}$. Taking m of order $T^{1/3}$ proves the theorem. Appendix B gives the full argument as a general integer-potential lemma, valid whenever at least r labels decrease a positive potential by one and no label increases it by more than one.

A single-depth argument also proves polynomial termination: reflecting at the global bound n yields $\mathbb{E}_\nu \tau = O_{k,r}(n^2(h + \log D + 1))$. The gain in (8) is its dependence on h in place of n , and the stronger tail below. For $h = \Theta(n)$ and fixed D , both expectation bounds have cubic order. The distance barrier uses an unknown witness. Below we replace it by an observable Hamming ball to obtain an efficient restart rule.

Theorem 3.2 (Sharp threshold and a gap from every single depth). *Let Φ_n contain all clauses with r positive and $k - r$ negative literals on distinct variables, with positive literals ordered first, and put $N = n - k + 2 \geq 3$.*

(i) *For every $0 < s < s_*$, some n , initial assignment, and single source in $\mathcal{Q}_{s,1}$ force nontermination under every nonanticipating clause selector.*

(ii) *At $s = s_*$, start with exactly r zero variables. Uniformly in $N \geq 3, T \geq 1$, the worst survival probability is*

$$A_{N,T} := \max_{\nu \in \mathcal{Q}_{s_*,1}} \mathbb{P}_\nu\{\tau > T\} = \exp\left[-\Theta_{k,r}\left(T^{1/3} + T/N^2\right)\right]. \quad (9)$$

For the same event $E_T = \{\tau > T\}$, let $F_t(E_T)$ be the minimum powered mass of a cover using only length- t cylinders. Then

$$\inf_{t \geq 0} F_t(E_T) = \Theta_{k,r}((T+1)^{-3/2}) \quad (1 \leq T \leq N-2). \quad (10)$$

Thus s_* is the sharp uniform threshold for this walk over the promise class. At a deadline proportional to n , variable-depth covers give $e^{-\Theta(n^{1/3})}$, while every single-depth cover of the same event costs at least order $n^{-3/2}$. The symmetric family supplies lower bounds; the upper theorem requires no such symmetry.

For $k = 3, r = 2$, the source condition is particularly simple: $\nu[w] \leq D2^{-3|w|/2}$. The distribution of the first t choices has min-entropy at least $1.5t - \log_2 D$ bits, with no conditional entropy bound on an individual choice. This promise problem is already tractable: replacing $(a \vee b \vee c)$ by the three pairwise clauses reduces finding a strong assignment to 2-SAT. The result concerns the robustness of the specified random walk, rather than a new tractability threshold.

The cubic dependence is necessary.

Theorem 3.3 (Sharp dependence on initial distance and source slack). *Fix $k/2 < r < k$ and let s_*, λ be as in (7). There is a constant $c_{k,r} > 0$ with the following property. For every integer $h \geq r$ and every $D \geq 1$, some complete formula Φ_n from Theorem 3.2 has a live initial assignment at Hamming distance h from its unique strongly r -satisfying assignment and a single source $\nu \in \mathcal{Q}_{s_*,D}$ such that, simultaneously for every clause selector,*

$$\mathbb{E}_\nu \tau \geq c_{k,r}(h + \log D + 1)^3.$$

The number of variables can be chosen to satisfy $n = O_{k,r}((h + \log D + 1)^3)$. Consequently the expectation bound in Theorem 3.1 is sharp in both h and D .

The lower bound uses the same formula family as the sharp tail. For fixed D and large h , one source can force a delay of order h^3 with probability one, simultaneously for every selector. Under the independent reference source, the expected number of flips is only $O_{k,r}(h)$.

An observable restart rule. The cover proof identifies excursions far from a strong solution as an obstruction to a better tail. Although the solution is unknown, a radius bound lets us cut off these excursions using only the current and initial assignments.

Theorem 3.4 (Restarting on leaving a Hamming ball). *Suppose the formula has a strongly r -satisfying assignment x^* at Hamming distance $h \geq 1$ from the initial assignment x_0 . Let $H \geq h$ be a supplied positive integer. Run the violated-clause walk, and reset to x_0 whenever an unsatisfying assignment leaves the Hamming ball of radius $2H$ around x_0 . Continue with the next unused source symbol. The rule uses H and x_0 , but does not use x^* . There are constants $c, C > 0$, depending only on k, r , such that for every clause selector, $D \geq 1$, $s \in [s_*, 1]$, and $\nu \in \mathcal{Q}_{s,D}$, its stopping time τ_H satisfies*

$$\mathbb{P}_\nu\{\tau_H > T\} \leq \min\{1, CD\lambda^h e^{-cT/H^2}\}, \quad T \in \mathbb{Z}_{\geq 0}. \quad (11)$$

In particular,

$$\mathbb{E}_\nu \tau_H = O_{k,r}\left(H^2(h + \log D + 1)\right).$$

The bound counts source-driven flips. A list of changed coordinates maintains the distance from x_0 in constant additional work per flip. Including resets, the total number of variable updates is at most twice the number of source-driven flips. Work to find a violated clause is separate.

The radius bound H is part of the guarantee: the algorithm needs neither the nearby witness nor its distance h , but it does need a valid upper bound on that distance. Taking $H = n$ is always valid but triggers no resets. A smaller supplied bound gives a tail scale H^2 , independently of the number of variables. No enumeration of assignments or capacity computation is required.

To see why the rule works, let y be an assignment just outside the ball. Then $\text{dist}(y, x^*) \geq 2H + 1 - h > h$. Resetting to x_0 therefore decreases the unknown witness distance. The restarted walk retains the favorable-flip comparison and keeps that distance at most $h + 2H$. A single-depth spectral estimate on this bounded range gives the exponential tail. Thus the benefit of the barrier analysis is an observable change to the algorithm; the changed algorithm then admits a simpler analysis. Appendix B gives the full proof.

For the complete formula family of Theorem 3.2, initialized with r zero variables, restarting after every unsuccessful flip already gives an exponential tail. Every favorable first flip satisfies the formula, so survival for T flips requires all T labels to lie in $\{r+1, \dots, k\}$. With $b = (k-r)k^{-s_*} = 1/(2\lambda) < 1$, the one-state cover recurrence gives $C_T = b^T$, and hence

$$\max_{\nu \in \mathcal{Q}_{s_*, D}} \mathbb{P}_\nu\{\tau_{\text{cutoff1}} > T\} = \min\{1, Db^T\}.$$

This family illustrates the benefit of restarting itself. Theorem 3.4 gives a guarantee for arbitrary strongly r -satisfiable formulas, where reaching a solution may require several flips. Proposition 4.3 separately distinguishes adaptive restarts from predetermined cutoffs.

4 When restarting restores exponential tails

A prefix cover also specifies an algorithmic decision: abandon an attempt when its observed word reaches the cover. The following criterion determines exactly when such decisions can restore exponential termination tails.

Fix an initial configuration x and a deterministic algorithm that consumes one source symbol per step. A restart resets the algorithm to x and resumes with the next unused symbol. The policy is deterministic, uses only observed symbols, and may restart at most once between successive steps. Running time counts source-consuming steps; reset costs are separate. Write $\mathcal{N}_x$ for the original nontermination event and $\alpha_s(x) = \mathcal{C}_s(\mathcal{N}_x)$, where $0 < s \leq 1$.

Theorem 4.1 (Exact restart criterion). *The following statements are equivalent:*

- (i) $\alpha_s(x) < 1$.
 - (ii) Some restart policy and constants $K < \infty$, $0 < r < 1$ satisfy $\Pr_\nu\{\tau_{\text{restart}} > t\} \leq DKr^t$ for every $D \geq 1$, $\nu \in \mathcal{Q}_{s,D}$, and integer $t \geq 0$.
 - (iii) Some restart policy terminates almost surely for every $\nu \in \mathcal{Q}_{s,1}$.
- If $\alpha_s(x) = 1$, every restart policy has nontermination probability one under some source in $\mathcal{Q}_{s,1}$.

Proof of sufficiency. If $\alpha_s(x) < 1$, continuity of finite-horizon capacities gives a horizon T and a finite prefix-free cover $\mathcal{B}$ of $\{\tau > T\}$, with words of length at most T , whose cost $c = \sum_{b \in \mathcal{B}} P[b]^s$ is less than one. Remove words on which termination has already occurred. Restart on reaching a word in $\mathcal{B}$. Every attempt ends within T steps. The words describing k failed attempts form the prefix-free family $\mathcal{B}^k$, of total cost c^k . Consequently,

$$\Pr_\nu\{\tau_{\text{restart}} > kT\} \leq D \sum_{w \in \mathcal{B}^k} P[w]^s = Dc^k.$$

This uses the source bound on whole execution prefixes; it requires no conditional entropy guarantee after a restart. The converse is proved in Appendix C. $\square$

The criterion has a stationary implementation when the live state space X is explicit. For a state update Φ , set $R(y, z) = \sum_{a: \Phi(y, a) = z} \pi(a)^s$, omitting terminated states.

Theorem 4.2 (Computing a restart rule). *The capacity vector $V(y) = \alpha_s(y)$ is the unique optimizer of*

$$\max \left\{ \sum_{y \in X} v(y) : 0 \leq v \leq \mathbf{1}, v \leq Rv \right\}. \quad (12)$$

If $V(x) < 1$, restarting whenever a transition enters $\{y : V(y) = 1\}$ gives an exponential tail uniformly over $\mathcal{Q}_{s,D}$ for every finite D . For rational entries of R , the rule is computable in polynomial time in the explicit state-space description.

The proof and quantitative bounds appear in Appendix C. An implicit state space can be exponentially large. The Hamming-ball rule of Theorem 3.4 avoids this enumeration by using the geometry of the assignments.

Proposition 4.3 (Adaptive versus predetermined restarts). *A three-live-state procedure has reference mean running time $8/3$ under a uniform four-symbol source and the following properties. For every predetermined sequence of positive integer cutoffs, allowing a final infinite cutoff, some source in $\mathcal{Q}_{1/2,1}$ gives nontermination probability at least $1/2$. In contrast, one stationary restart rule satisfies, for every $D \geq 1$ and integer $T \geq 2$,*

$$\max_{\nu \in \mathcal{Q}_{1/2,D}} \Pr_\nu\{\tau_{\text{adaptive}} > T\} = \min\{1, DT2^{1-T}\}.$$

Restart policies based on elapsed time and observations are established. Luby, Sinclair, and Zuckerman study universal schedules for independent runs [LSZ93]; dynamic policies and models of dependence among runs are studied in [KHR⁺02, RHK02]. Here the capacity criterion and the separation concern one global prefix constraint, which supplies no fresh randomness at a reset. The separating example is given in Appendix C.

5 An optimal order invisible to stopping-time analysis

Under independent randomness, changing the order of independent retry tasks cannot change the distribution of their total running time. Under a global prefix constraint, the order can determine whether a linear deadline is missed with probability one or exponentially small probability. We first identify an optimal order for arbitrary heterogeneous tasks, then quantify this distinction on two task types.

There are n unfinished tasks and a finite symbol alphabet $\mathcal{A}$. Task i has a nonempty success set $B_i \subseteq \mathcal{A}$. An attempt chooses an unfinished task and reads the next symbol. The task completes if the symbol lies in B_i ; otherwise it remains unfinished. Completed tasks are never selected again, and updating one task does not change another. The success sets may overlap. A deterministic nonanticipating selector may use the entire observed history. Let τ count attempts until all tasks complete. For $P = \pi^{\mathbb{N}}$ with full-support π , put

$$W = \sum_{a \in \mathcal{A}} \pi(a)^s, \quad w_i = \sum_{a \in B_i} \pi(a)^s.$$

Theorem 5.1 (Optimal priority under a prefix constraint). *For every $0 < s \leq 1$, always attempting an unfinished task with smallest w_i minimizes $\max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_{\nu}\{\tau > T\}$ over all deterministic nonanticipating selectors, simultaneously for every $T \geq 0$ and $D \geq 1$. If $w_1 \leq \dots \leq w_n$, set $v_t(n+1) = 0$, $v_0(j) = 1$ for $1 \leq j \leq n$, and*

$$v_{t+1}(j) = \min\{1, (W - w_j)v_t(j) + w_j v_t(j+1)\}.$$

The exact optimum is $\min\{1, Dv_T(1)\}$. Once the masses are computed and sorted, this takes $O(nT)$ arithmetic operations and $O(n)$ memory. Nevertheless, every selector has the same stopping-time law under P and the same single-depth prefix power sums for every positive power.

The rule completes tasks in increasing order of powered success mass; switching among unfinished tasks cannot improve the guarantee. For singleton success sets, or for a uniform reference source, this is increasing order of ordinary success probability, independently of s . In general the order can depend on s . Appendix D.2 proves a weighted exchange inequality that survives the truncation at one in the prefix recursion. It reduces an apparent dynamic program on 2^n subsets to n suffixes.

A quantitative separation. Take n tasks of type A , successful on 0, and one task of type B , successful on 1. Write Λ_A for completing all A tasks first and Λ_B for completing B first.

Theorem 5.2 (Equal prefix sums, different robustness). *Fix $P(0) = 15/16$, $P(1) = 1/16$, and $\mathcal{Q} = \mathcal{Q}_{1/2,1}$. For every $n \geq 1$,*

$$\max_{\nu \in \mathcal{Q}} \mathbb{P}_{\nu}^{\Lambda_A}\{\tau > 8n\} = 1, \tag{13}$$

$$\left(\frac{15}{16}\right)^{4n} \leq \max_{\nu \in \mathcal{Q}} \mathbb{P}_{\nu}^{\Lambda_B}\{\tau > 8n\} \leq \frac{9}{8} \left(\frac{225}{256}\right)^n < 1. \tag{14}$$

One source $\nu_n \in \mathcal{Q}$ gives survival probability one for Λ_A and at most the upper bound in (14) for Λ_B . All deterministic nonanticipating selectors have identical prefix power sums at every depth and every positive power, and identical stopping-time laws under P . The lower bound holds for every selector; moreover, Λ_B is exactly minimax optimal at every deadline by Theorem 5.1.

Why all power sums agree. For $q > 0$, let $W_q = \sum_a \pi(a)^q$ and $\pi_q(a) = \pi(a)^q/W_q$. The surviving-prefix sum is

$$\tilde{Z}_T^\Lambda(q) = W_q^T \mathbb{P}_{\pi_q^\Lambda}^{\Lambda} \{\tau > T\}.$$

Under this tilted independent law, preassign an independent stream to each task. The total number of attempts is a sum of independent geometric workloads, regardless of the selector. This proves equality for every q , including history-dependent selection. The sums that include a terminating T th attempt agree as well: $Z_0(q) = 1$ and $Z_T(q) = W_q \tilde{Z}_{T-1}(q)$ for $T \geq 1$.

Why the robust tails differ. To defeat Λ_A , generate zeros with probability $3/4$ until the n th zero, then force zeros through time $8n$ and use P afterwards. The initially random segment makes the later deterministic segment admissible under the global prefix constraint. Task B receives no successful symbol before the deadline. For Λ_B , cover first by the prefix 0^{4n} . Otherwise the first 1 completes B by time $4n$, and survival requires fewer than n zeros in the remaining symbols. A generating-function bound covers these remaining paths at their full depth. The resulting mixed-depth cover proves (14); Appendix D.3 checks the source constraint and both bounds.

Thus the cover changes an algorithmic choice that every single-depth power sum leaves unresolved. One direct use is obtaining an accepted proposal for each of several rejection tasks: finish the task with smallest powered acceptance mass first. Those masses must be supplied or efficiently computable. This is a termination guarantee; correlated proposals need not preserve the output distribution of independent rejection sampling.

6 Composition under a shared source

Sharing a random stream can alter robustness even when a selector is free to interleave the components. The following example makes the distinction between a global source guarantee and separate component guarantees exact.

Each component is a connected N -vertex graph with binary vertex values. Select an edge whose endpoints disagree. Four source labels replace its endpoint values by 00, leave them unchanged, reverse them, or replace them by 11. The reference probabilities are $(49/100, 1/4, 1/4, 1/100)$, respectively. Constant components are finished and never selected again. This extends the discordant voting model of Cooper, Dyer, Frieze, and Rivera [CDFR18].

Let $\Delta_N \in (1/2, 1)$ be the unique solution of

$$2 \cdot 4^{-\Delta_N} + 2(7/100)^{\Delta_N} \cos(\pi/N) = 1.$$

We evaluate robustness at this exponent; Appendix E gives the exact formula.

Corollary 6.1 (Small individual failure, certain joint failure). *For every $N \geq 3$, start each of $N - 1$ components with one vertex equal to one. In isolation, each has worst-case nontermination probability*

$$\max_{\nu \in \mathcal{Q}_{\Delta_N, 1}} \nu\{\tau_i = \infty\} = 7^{-\Delta_N(N-2)} = \exp[-\Theta(N)].$$

There is one source in $\mathcal{Q}_{\Delta_N, 1}$ under which their combined execution never terminates, simultaneously for every deterministic nonanticipating rule that chooses a disagreeing edge. The rule may interleave components arbitrarily and remember all past labels. Under the independent reference source, every such rule has expected total running time at most $25(N - 1)/12$.

The proof counts the ones in components that have not yet finished. Below $N - 1$, no component can finish by becoming all ones, and a harmonic potential bounds the cost of reaching that level. Starting from $N - 1$, a single admissible source keeps the count between 1 and $N - 1$, under every interleaving. The source suppresses downward moves at 1 and upward moves at $N - 1$, so no component can finish by becoming all ones and the last unresolved one can never disappear. The count need not determine the full transition law. Theorem E.1 gives the exact formula for arbitrary initial counts and components of size at least N ; Appendix E proves it.

The global prefix bound need not hold on a component’s interleaved subsequence. The corollary therefore identifies a limitation of a shared randomness assumption, rather than contradicting composition theorems based on separate componentwise assumptions. The same source feature explains why the restart proof must use concatenated whole prefixes instead of reapplying a conditional guarantee after each reset.

7 Context and further questions

The source assumption belongs to the study of weak randomness [CG88, DMOZ23]. Recent work studies Markov-chain mixing and hitting times under perturbations of each transition [Zuc26], and online condensing of streams that may contain predictable steps [DMOZ25]. The latter model includes general high-min-entropy sources; it does not require fresh entropy at every step. Here the constraint is imposed on complete execution prefixes, and the objective is a sharp termination guarantee or a control rule for the specified process.

The promise-SAT walk and its ordinary Hamming-distance analysis are classical [AGH17]. The robustness theorem uses only an integer progress comparison, recorded in Lemma B.1. The cube-root scale has a direct antecedent in boundary-case branching random walks. A comparison with Mallein’s consistent-displacement theorem [Mal19, Theorem 1.1] implies even the probability-one plateau used in our distance lower bound. Lemma B.2 gives an elementary proof. Our contribution here is its interpretation as a sharp delay under a global prefix constraint and its realization by one SAT source that works for every clause selector. Search within Hamming balls is established in SAT algorithms [DHW03]. Our restart result concerns the robustness gain from a same-center ball-exit rule under one cumulative prefix constraint; it does not supply a new SAT tractability threshold.

Tree max-flow/min-cut and the relations between weighted growth and dimension are established [BP17, Sta93, MW88, Sim15]. Achlioptas, Iliopoulos, and Vlassis already use cuts at different depths of trajectory trees in entropy compression [AIV17]. The capped recursion can also be viewed as multiplicative optimal stopping after normalizing powered transition weights [JPS21]. Our restart and priority theorems give algorithmic consequences of the exact source interpretation; the scheduling example shows why all single-depth power sums together are insufficient.

The investigation of hierarchical covers was motivated by Li and Xu’s construction of local subspace priors and a prior on the Grassmannian [LX26, Section 3.3]. Xu’s Bellman-sufficient formulation studies the information needed to close a controlled recursion [Xu26]. Here the retained information is the nesting of execution prefixes, and the recursion optimizes robustness under a global tape constraint.

Can the Hamming-ball radius be chosen adaptively, with comparable guarantees under one global source constraint? Which other observable progress measures give useful restart rules? A further question is to identify composition rules that preserve robustness under interleaving.

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A Prefix covers and worst-case sources

Throughout this appendix, $P = \pi^{\mathbb{N}}$ is the full-support finite source from Section 2. The source class $\mathcal{Q}_{s,D}$ imposes its bound on every prefix of the full tape. Symbols after termination may be continued with the reference law.

Proof of Theorem 2.1. Every prefix cover gives $\nu(E) \leq D \sum_{w \in \mathcal{B}} P[w]^s$, which proves the upper bound. Suppose first that E is determined by T symbols. In the full source tree through depth T , give the root capacity $c(\emptyset) = 1$ and every other node w capacity $c(w) = DP[w]^s$. Route flow from the root only into the depth- T words whose cylinders lie in E . The tree max-flow/min-cut theorem gives a largest admissible flow f of value $\min\{1, DC_s(E)\}$. Indeed, cuts are exactly prefix covers, with root cost 1; a cover of a whole depth- T cylinder by longer words cannot lower its cost, since for nonnegative x_i and $s \leq 1$, $\sum_i x_i^s \geq (\sum_i x_i)^s$.

It remains to extend f to a probability law on all depth- T words. Extend f by zero on branches outside its support. For every internal node, including the root,

$$\sum_a c(wa) \geq c(w).$$

For nonroot nodes this follows from $\sum_a \pi(a)^s \geq 1$; at the root it also uses $D \geq 1$. Since f conserves flow, the residual capacities $r(w) = c(w) - f(w)$ satisfy

$$\sum_a r(wa) \geq r(w).$$

Start an additional flow g of value $1 - f(\emptyset)$ at the root. Whenever its incoming mass at w is at most $r(w)$, distribute that mass among its children without exceeding their residual capacities. The displayed inequality makes this possible at every step. Then $f + g$ has total mass one and respects all node capacities. It cannot give E more than the maximal value of f , because its restriction to successful leaves would itself be a larger admissible flow. Thus this probability law attains the upper bound. Continue it beyond depth T using the reference conditional law. For any suffix v ,

$$\nu[wv] = \nu[w]P[v] \leq DP[w]^s P[v]^s = DP[wv]^s,$$

so it belongs to $\mathcal{Q}_{s,D}$ on the infinite tape space.

For a closed E , let E_n be the union of the length- n cylinders that intersect E . These are decreasing closed-and-open sets with intersection E . Their contents decrease to $\mathcal{C}_s(E)$: monotonicity gives one inequality, while any finite prefix cover of E covers all sufficiently large E_n , by compactness. The class $\mathcal{Q}_{s,D}$ is nonempty and weakly compact. Indeed, $P \in \mathcal{Q}_{s,D}$, the tape space is compact, and each cylinder constraint is closed under weak convergence. Choose maximizers ν_n for E_n and a weakly convergent subsequence with limit $\nu_* \in \mathcal{Q}_{s,D}$. For fixed m and $n \geq m$,

$$\nu_n(E_m) \geq \nu_n(E_n) = \min\{1, DC_s(E_n)\}.$$

Pass to the subsequential limit, using that E_m is closed and open, and then let $m \rightarrow \infty$. Continuity from above yields $\nu_*(E) \geq \min\{1, DC_s(E)\}$, proving attainment and (3). For any decreasing sequence of finite-symbol events E_T with intersection $\mathcal{N}$, the same compactness argument shows $\mathcal{C}_s(E_T) \downarrow \mathcal{C}_s(\mathcal{N})$ and proves (4). $\square$

Corollary A.1 (Dimension as a source threshold). *Let $\mathcal{N}$ be closed and put $\delta = \dim_H^P \mathcal{N}$. For $0 < s < \delta$, some finite D and some $\nu \in \mathcal{Q}_{s,D}$ satisfy $\nu(\mathcal{N}) = 1$. For $\delta < s \leq 1$, every such law gives $\nu(\mathcal{N}) = 0$. For every $0 < s \leq 1$, the least domination constant permitting a law supported on $\mathcal{N}$ is*

$$\inf\{D \geq 1 : \exists \nu \in \mathcal{Q}_{s,D}, \nu(\mathcal{N}) = 1\} = \frac{1}{\mathcal{C}_s(\mathcal{N})}, \quad (15)$$

where $1/0 = \infty$. If $P(\mathcal{N}) = 0$, then

$$\delta = \inf\{s \in (0, 1] : \nu(\mathcal{N}) = 0 \text{ for every finite } D \geq 1 \text{ and every } \nu \in \mathcal{Q}_{s,D}\}. \quad (16)$$

At $s = \delta > 0$, there exists a law with finite domination constant supported on $\mathcal{N}$ exactly when $\mathcal{C}_\delta(\mathcal{N}) > 0$; dimension alone does not settle this endpoint.

Proof of Corollary A.1. In the metric $d_P(\omega, \eta) = P[\omega \wedge \eta]$, each cylinder $[w]$ has diameter $P[w]$. Any nonsingleton set is contained in a cylinder with the same diameter; singleton sets can be covered by cylinders of arbitrarily small diameter. Thus the infimum over countable cylinder covers equals the Hausdorff content. For compact $\mathcal{N}$, every such cover has a finite subcover, so this infimum is $\mathcal{C}_s(\mathcal{N})$. Its zero threshold is therefore $\dim_H^P \mathcal{N}$: the content is positive for $s < \delta$ and zero for $s > \delta$. Theorem 2.1 proves (15) and the strict-threshold claims. If $P(\mathcal{N}) = 0$, then $\mathcal{C}_1(\mathcal{N}) = P(\mathcal{N}) = 0$, so the set in (16) is nonempty and has infimum δ . The same exact identity gives the stated critical criterion. $\square$

B Proofs for the promise-SAT walk

B.1 A barrier bound from an integer potential

The upper bound uses only a property of the potential after each possible source symbol. In particular, it needs neither a finite state space nor a Markov transition law for the potential.

Lemma B.1 (Integer-potential barrier). *Fix $k/2 < r < k$ and the parameters λ, s_* in (7). Consider a deterministic process driven by symbols in $[k]$, with uniform product reference law P and stopping time τ . Assign to every live history a positive integer potential d , initially $h \geq 1$, and give every terminated history potential zero. At each live history, assume that at least r of the k next symbols either terminate or give a live potential at most $d - 1$, and that every other symbol either terminates or gives a live potential at most $d + 1$. For every integer $m > h$, every $T \geq 0$, every $D \geq 1$, and every $\nu \in \mathcal{Q}_{s_*,D}$,*

$$\mathbb{P}_\nu\{\tau > T\} \leq \min \left\{ 1, D \left[\frac{h}{m} \lambda^{h-m} + \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \exp\left(-\frac{\pi^2 T}{2(2m-1)^2}\right) \right] \right\}. \quad (17)$$

Consequently (8) holds, with constants depending only on k, r .

Proof. Put

$$a = rk^{-s_*} = \lambda/2, \quad b = (k-r)k^{-s_*} = 1/(2\lambda).$$

If f is nonnegative and increasing, with $f(0) = 0$, the sum of f over the live children of a history of potential d , multiplied by the powered edge weight k^{-s_*} , is at most

$$af(d-1) + bf(d+1). \quad (18)$$

Indeed, assign r of the decreasing symbols to the first term. Each other live child has potential at most $d + 1$; terminated children contribute zero. This argument applies separately after every history.

Cover $\{\tau > T\}$ by prefixes that first reach potential m while still live, up to time T , and live length- T prefixes that never reach m . Remove any length- T prefix already in the first set. Because upward increments are at most one, the first set meets every path that leaves $\{1, \dots, m - 1\}$. The resulting cover is prefix-free.

For first-hit prefixes use

$$H_m(d) = \frac{d}{m} \lambda^{d-m} \quad (0 \leq d \leq m).$$

This function is increasing, $H_m(0) = 0$, $H_m(m) = 1$, and

$$aH_m(d-1) + bH_m(d+1) = H_m(d) \quad (1 \leq d < m).$$

Start with weight $H_m(h)$ at the root. Expanding a live node below m cannot increase the sum of its powered prefix weight times H_m , by (18). Leave first-hit nodes unexpanded, give terminated nodes value zero, and continue to depth T . Every first-hit node has value one; the remaining values are nonnegative. Finite-tree induction therefore bounds the powered mass of first-hit prefixes by $H_m(h)$.

For paths remaining below m , put $L = m - 1$ and define a reflected comparison matrix $R^{(m)}$ on $\{1, \dots, L\}$ by

$$(R^{(m)}f)(d) = af(d-1) + bf(d+1) \quad (d < L), \quad (R^{(m)}f)(L) = af(L-1) + bf(L),$$

where $f(0) = 0$. This matrix preserves nonnegative increasing vectors: consecutive differences in the interior are nonnegative linear combinations of consecutive differences of f , and the final difference is $a(f(L-1) - f(L-2)) \geq 0$ when $L \geq 2$. The claim is immediate for $L = 1$. Thus $F_t = (R^{(m)})^t \mathbf{1}$ is nonnegative and increasing. Induction on the remaining horizon bounds the powered mass of length- T live paths that stay below m by $F_T(h)$. For a history with potential below L , apply (18) to F_t . At potential L , an upward exit is discarded, each of r decreasing symbols contributes at most $F_t(L-1)$, and any other retained child contributes at most $F_t(L)$. This gives the reflected final row.

Let $B = \text{diag}(\lambda, \dots, \lambda^L)$. The symmetric matrix $B^{-1}R^{(m)}B$ has off-diagonal entries $1/2$, last diagonal entry $b < 1/2$, and other diagonal entries zero. Increasing the last diagonal entry to $1/2$ gives a nonnegative symmetric matrix with largest eigenvalue $\cos(\pi/(2L+1))$. To see the latter, the positive vector $v_j = \sin(j\pi/(2L+1))$ satisfies its eigenvector equation, including the reflected final row. A symmetric nonnegative matrix has Euclidean operator norm equal to its spectral radius. Perron–Frobenius monotonicity consequently gives

$$\|B^{-1}R^{(m)}B\|_2 \leq \cos \frac{\pi}{2m-1}.$$

Since $\|(\lambda^{-j})_{j=1}^L\|_2 \leq 1/\sqrt{\lambda^2 - 1}$,

$$F_T(h) \leq \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \left(\cos \frac{\pi}{2m-1} \right)^T \leq \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \exp \left(-\frac{\pi^2 T}{2(2m-1)^2} \right).$$

The source constraint bounds the probability of each cover cylinder by D times its powered reference mass. Summing the two parts proves (17).

When $T^{1/3} > h$, choose an integer $m > h$ within two of $T^{1/3}$. Both terms are at most $C\lambda^h e^{-cT^{1/3}}$. When $T^{1/3} \leq h + 1$, the same capped bound follows from probability at most one, after enlarging C and decreasing c . Finally, summing $\min\{1, CD\lambda^h e^{-cT^{1/3}}\}$ over integer $T \geq 0$ gives $C'(h + \log D + 1)^3$. For example, split the sum at a sufficiently large constant times $(h + \log D + 1)^3$; beyond that point its summands are bounded by $e^{-cT^{1/3}/2}$. $\square$

Proof of Theorem 3.1. Let d be the Hamming distance to x^* . Whenever a clause is violated, every literal it satisfies under x^* belongs to a coordinate on which the current assignment and x^* disagree. Therefore at least r flips decrease d by one; all other flips increase it by one. If the formula becomes satisfied, terminate and assign potential zero. Every live assignment has positive distance from x^* . Lemma B.1 applies for every clause selector and proves the claim. If the initial assignment is already satisfying, $\tau = 0$. Finally, $\mathcal{Q}_{s,D} \subseteq \mathcal{Q}_{s^*,D}$ for $s \geq s^*$. Under P , the expected decrease in distance at any live history is at least $(2r - k)/k$; the additive-drift bound gives $\mathbb{E}_P \tau \leq kh/(2r - k)$. $\square$

The single-depth baseline. For SAT, live distances are at most n . Use the reflected matrix above on $\{1, \dots, n\}$. At distance n , every actual flip decreases distance, so its powered contribution on an increasing vector f is at most $(a + b)f(n - 1) \leq af(n - 1) + bf(n)$. The same induction and spectral estimate therefore bound the single-depth live-prefix sum by

$$\tilde{Z}_T(s_*) := \sum_{w \in [k]^T : \tau(w) > T} k^{-s_* T} \leq \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \exp\left(-\frac{\pi^2 T}{2(2n + 1)^2}\right). \quad (19)$$

Applying the source bound at depth T and summing the capped tail proves $\mathbb{E}_\nu \tau = O_{k,r}(n^2(h + \log D + 1))$.

B.2 A family with a sharp source threshold

For $n \geq k + 1$, let Φ_n contain every clause

$$\left(\bigvee_{i \in I} x_i\right) \vee \left(\bigvee_{j \in J} \neg x_j\right), \quad I \cap J = \emptyset, \quad |I| = r, \quad |J| = k - r. \quad (20)$$

There are $\binom{n}{k} \binom{k}{r} = O_k(n^k)$ clauses. The all-one assignment strongly r -satisfies the formula. It is the unique such assignment. For an assignment with $z \geq 1$ zeroes and $o = n - z$ ones, if $o \geq k - r$, put $k - r$ ones in the negative positions and at least one zero in the positive positions; the clause has fewer than r true literals. If $o < k - r$, put all the ones in negative positions and fill the remaining positions with zeroes; the clause has $k - r - o < r$ true literals. The required choices exist because $n \geq k$.

Order positive literals first in each clause. If d is the number of zero variables, a clause is violated precisely when its positive variables are zero and its negative variables are one. Hence an assignment is live exactly when $r \leq d \leq n - k + r$. Put $N = n - k + 2$ and $j = d - r + 1$; the live values are $1, \dots, N - 1$. For every selector, symbols $1, \dots, r$ decrease j , and the other symbols increase it. The powered matrix of live transitions is

$$R_s(j, j - 1) = rk^{-s}, \quad R_s(j, j + 1) = (k - r)k^{-s}, \quad (21)$$

with out-of-range entries omitted.

Proof of Theorem 3.2(i). The spectral radius of (21) is

$$\rho_s = 2\sqrt{r(k-r)} k^{-s} \cos(\pi/N).$$

For $s < s_*$, take N sufficiently large that $\rho_s > 1$. A positive right eigenvector is $\varphi(j) = \lambda^j \sin(j\pi/N)$. Start at a maximizing j_0 , and at live value j give label a probability

$$q(a | j) = k^{-s} \frac{\varphi(j + \delta(a))}{\rho_s \varphi(j)}, \quad \delta(a) = \begin{cases} -1, & a \leq r, \\ 1, & a > r, \end{cases}$$

where $\varphi(0) = \varphi(N) = 0$. The eigenvector equation normalizes these probabilities, and the boundary labels have probability zero. For every supported word w of length t , telescoping gives

$$\nu[w] = k^{-st} \rho_s^{-t} \frac{\varphi(j_t)}{\varphi(j_0)} \leq k^{-st} = P[w]^s.$$

The resulting source belongs to $\mathcal{Q}_{s,1}$ and never exits the live interval. The path j_t is determined by source labels alone, independently of the clause selector. Thus this one source forces nontermination for all selectors simultaneously. $\square$

B.3 Matching tails and the gap from single-depth covers

Proof of Theorem 3.2(ii). For brevity write $R = R_{s_*}$ and $a = \lambda/2$, $b = 1/(2\lambda)$. The exact distance reduction gives the prefix-capacity recursion

$$C_0 = \mathbf{1}, \quad C_{t+1} = \min\{\mathbf{1}, RC_t\},$$

with the minimum taken coordinatewise. The prefix-cover identity implies $A_{N,T} = C_T(\mathbf{1})$. Theorem 3.1, with $h = r$, gives $A_{N,T} \leq C_1 e^{-c_1 T^{1/3}}$. Also, diagonal similarity transforms R to the symmetric killed simple-walk matrix on $1, \dots, N-1$. Its Euclidean norm is $\cos(\pi/N)$, so the full-depth cover gives

$$A_{N,T} \leq e_1^\top R^T \mathbf{1} \leq \frac{\lambda}{\sqrt{\lambda^2 - 1}} e^{-\pi^2 T / (2N^2)}.$$

Taking the smaller bound gives a constant times $\exp[-c_2(T^{1/3} + T/N^2)]$. Since $A_{N,T} \leq A_{N,1} = b < 1$, decreasing c_2 absorbs the leading constant uniformly for all $N \geq 3, T \geq 1$.

For a lower bound, retain only paths within $1, \dots, m-1$, where $3 \leq m \leq N$, and define

$$\rho_m = \cos(\pi/m), \quad \varphi_m(j) = \lambda^j \sin(j\pi/m), \quad u(j) = \frac{\varphi_m(j)}{\max_{1 \leq i < m} \varphi_m(i)}.$$

Extend u by zero outside this interval. Then $0 \leq u \leq 1$ and $Ru \geq \rho_m u$. Induction in the capped recursion gives $C_T \geq \rho_m^T u$: if this holds at T , then $RC_T \geq \rho_m^{T+1} u$, and capping at one preserves the inequality. Furthermore

$$u(1) \geq \frac{2}{m} \lambda^{2-m}, \quad \rho_m^T \geq e^{-\pi^2 T / m^2} \quad (m \geq 3).$$

Consequently

$$A_{N,T} \geq \frac{2}{m} \lambda^{2-m} e^{-\pi^2 T / m^2}.$$

Choose $m = \min\{N, \max\{3, \lceil T^{1/3} \rceil\}\}$. For $T \leq N^3$ this is at least $e^{-CT^{1/3}}$; for $T > N^3$ it is at least $e^{-C(N+T/N^2)} \geq e^{-2CT/N^2}$. This proves (9).

For the single-depth comparison, write S_N for the symmetric killed simple-walk matrix. Diagonal similarity gives

$$e_1^\top R^t \mathbf{1} = \sum_{j=1}^{N-1} \lambda^{1-j} S_N^t(1, j).$$

The reflection principle on the positive half-line gives, when parities match,

$$S_\infty^t(1, j) = 2^{-t} \left[\binom{t}{(t+j-1)/2} - \binom{t}{(t+j+1)/2} \right] \leq C \frac{j}{(t+1)^{3/2}}.$$

For the inequality, factor the binomial difference as $2j/(t+j+1)$ times its first term and use the central-binomial bound. Since $S_N^t \leq S_\infty^t$ and $\sum_{j \geq 1} j \lambda^{1-j} < \infty$, the single-depth sum is at most $C(t+1)^{-3/2}$.

For even $t = 2\ell \leq N-2$, the Catalan paths from 1 back to 1 stay below N . Their powered mass is $\text{Cat}_\ell(ab)^\ell = \text{Cat}_\ell / 4^\ell \geq c(t+1)^{-3/2}$. For odd $t = 2\ell + 1 \leq N-2$, append one upward step to these paths; this multiplies the mass by b and gives the same order. Thus the single-depth sum is $\Theta_{k,r}((t+1)^{-3/2})$ for every $0 \leq t \leq N-2$.

It remains to compare covers of the same event $E_T = \{\tau > T\}$ at all depths, including depths exceeding T . Every live state has a live neighbor when $N \geq 3$, and hence has an arbitrarily long live continuation. For $t \leq T$, the length- t cylinders meeting E_T are exactly the live words. Therefore $F_t(E_T) = e_1^\top R^t \mathbf{1}$. For $t > T$, all k^{t-T} extensions of each live length- T word belong to E_T , so

$$F_t(E_T) = k^{(1-s_*)(t-T)} e_1^\top R^T \mathbf{1} \geq F_T(E_T).$$

Taking the infimum over all depths proves (10). □

B.4 Sharp dependence on distance and source slack

The initial-distance lower bound uses a feature that a fixed-strip eigenvector does not capture. The capped recurrence retains a region where its value is exactly one. The boundary of this region advances no faster than order $t^{1/3}$.

Lemma B.2 (A slowly advancing plateau). *Let $\lambda > 1$, $a = \lambda/2$, and $b = 1/(2\lambda)$. On the positive integers define*

$$C_0(j) = 1, \quad C_t(0) = 0, \quad C_{t+1}(j) = \min\{1, aC_t(j-1) + bC_t(j+1)\}.$$

There are constants $m_0, K > 0$, depending only on λ , such that

$$C_t(j) = 1 \quad \text{whenever} \quad j \geq (m_0^3 + Kt)^{1/3}.$$

The order of this plateau also follows from Mallein [Mal19, Theorem 1.1]. To see the comparison, put $\beta = \log \lambda$. Take a branching random walk with bounded offspring, at least one child per particle, and expected child counts a, b at respective displacements $+\beta, -\beta$. Its boundary identities are $ae^{-\beta} + be^\beta = 1$ and $a\beta e^{-\beta} - b\beta e^\beta = 0$. The probability $q_t(j)$ of a lineage staying below $j\beta$ through generation t satisfies $q_t(j) \leq C_t(j)$ by the union bound. Mallein's theorem gives $q_t(\lceil At^{1/3} \rceil \pm 1) \rightarrow 1$ for sufficiently large A . Since $a + b > 1$, the next capped step equals one. Monotonicity and larger constants cover all t . We give a direct proof for completeness.

Proof. Write $\beta = \log \lambda$. For each sufficiently large real m , let $\theta_m \in (0, \pi/m)$ be the unique solution of

$$m\theta_m + \arctan(\theta_m/\beta) = \pi.$$

Define a function on $[0, \infty)$ by

$$f_m(x) = \begin{cases} e^{\beta(x-m)} \frac{\sin(\theta_m x)}{\sin(\theta_m m)}, & 0 \leq x < m, \\ 1, & x \geq m. \end{cases}$$

The function increases from zero to one. Indeed, its logarithmic derivative below m is $\beta + \theta_m \cot(\theta_m x) > 0$, and this derivative vanishes at m . Write g_m for the sine expression without truncation and $\rho_m = \cos \theta_m$. Direct calculation gives

$$ag_m(x-1) + bg_m(x+1) = \rho_m g_m(x).$$

For $0 \leq z \leq 1$,

$$g_m(m+z) = e^{\beta z} \left(\cos(\theta_m z) - \frac{\beta}{\theta_m} \sin(\theta_m z) \right) \leq 1;$$

its derivative in z is $-e^{\beta z}(\beta^2/\theta_m + \theta_m) \sin(\theta_m z) \leq 0$. Thus replacing g_m by f_m cannot decrease a neighbor value used at an integer $j < m$. If $Ff(j) = \min\{1, af(j-1) + bf(j+1)\}$, then

$$Ff_m(j) \geq \rho_m f_m(j) \quad (1 \leq j < m). \quad (22)$$

There is also a fixed $z_0 \in (0, 1)$ such that, for all sufficiently large m ,

$$Ff_m(j) = 1 \quad (j \geq m - z_0). \quad (23)$$

To see this, for every fixed $z \geq 0$,

$$f_m(m-z) \longrightarrow e^{-\beta z}(1 + \beta z).$$

At $z_0 = 0$ the limiting value of $af_m(m - z_0 - 1) + b$ is

$$\frac{1 + \beta + e^{-\beta}}{2} > 1.$$

Continuity permits a sufficiently small positive z_0 with the same strict inequality. Monotonicity of f_m , together with $f_m(j+1) = 1$ when $j \geq m - z_0$, proves (23).

Away from this saturated band, increasing m decreases f_m at a uniform logarithmic rate. Differentiating the defining equation for θ_m gives

$$-\frac{\theta'_m}{\theta_m} = \frac{1}{m + \beta/(\beta^2 + \theta_m^2)}.$$

For fixed $0 < x < m$ it follows that

$$-\partial_m \log f_m(x) = \frac{\theta_m x \cot(\theta_m x) + \beta m}{m + \beta/(\beta^2 + \theta_m^2)}. \quad (24)$$

The function $y \cot y$ decreases on $(0, \pi)$, so the right side is bounded below, for $x \leq m - z_0$, by its value at $x = m - z_0$. That value converges to $\beta^2 z_0 / (1 + \beta z_0) > 0$. Hence there are m_0 and $\eta > 0$ such that $-\partial_m \log f_m(x) \geq \eta$ whenever $m \geq m_0$ and $0 < x \leq m - z_0$.

Set

$$\delta_m = \frac{-\log \rho_m}{\eta}.$$

For $j < m - z_0$, integration of (24) over $[m, m + \delta_m]$ yields $f_{m+\delta_m}(j) \leq \rho_m f_m(j) \leq F f_m(j)$. For the remaining j , use (23). We have proved the pointwise comparison

$$F f_m \geq f_{m+\delta_m}.$$

Since $\theta_m < \pi/m$, increasing m_0 if necessary gives $\delta_m \leq A/m^2$ for a constant A . Starting from m_0 , define $m_{t+1} = m_t + \delta_{m_t}$. Monotonicity of F and $C_0 \geq f_{m_0}$ give $C_t \geq f_{m_t}$ by induction. Moreover

$$m_{t+1}^3 - m_t^3 \leq 3A + 3A^2/m_0^3 + A^3/m_0^6 =: K.$$

Thus $m_t \leq (m_0^3 + Kt)^{1/3}$, and $C_t(j) = 1$ for every $j \geq (m_0^3 + Kt)^{1/3}$. $\square$

Proof of Theorem 3.3. Put $j = h - r + 1$. The source-label sequence determines the distance process of Φ_n independently of the clause selector. For a horizon T , choose $N > j + T$ and $n = N + k - 2$. No path starting at j can reach the upper boundary by time T , so the finite-interval capacity at (j, T) equals the half-line value $C_T(j)$ of Lemma B.2. When j is sufficiently large, this value is one for $T = \lfloor c_1 j^3 \rfloor$, with a fixed $c_1 > 0$. The finite-tree capacity identity therefore supplies a source in $\mathcal{Q}_{s_*, 1}$ under which $\tau > T$ with probability one. The source may be extended beyond depth T by independent uniform symbols; because $s_* < 1$, this preserves every prefix bound.

For completeness, source slack has the same cubic effect. The finite-horizon capacity identity, with root capacity one and all nonroot capacities multiplied by D , gives

$$\sup_{\nu \in \mathcal{Q}_{s_*, D}} \Pr_{\nu}\{\tau > T\} = \min\{1, DC_T(j)\}.$$

The half-line recurrence is increasing in j . The eigenvector lower bound used in Theorem 3.2(ii), with a strip of width $\max\{3, \lceil T^{1/3} \rceil\}$, gives $C_T(1) \geq \exp[-B(T+1)^{1/3}]$ for a constant B . For sufficiently large $\log D$, choose $T = \lfloor c_2 (\log D)^3 \rfloor$ with sufficiently small $c_2 > 0$. Then $DC_T(j) \geq DC_T(1) \geq 1$. Consequently a source in $\mathcal{Q}_{s_*, D}$ forces survival beyond this horizon with probability one.

Choose whichever of the two horizons is larger, and then take $N = j + T + 2$. This gives $\mathbb{E}_{\nu} \tau \geq c(h + \log D + 1)^3$ after reducing the constant: for bounded j and $\log D$, the live initial assignment already implies $\tau \geq 1$. The same construction has $n = O((h + \log D + 1)^3)$. All claims hold for every selector under the same source because the label-to-distance map is selector independent. $\square$

B.5 An observable restart rule for the promise-SAT walk

Proof of Theorem 3.4. Use the unknown witness distance $d(x) = \text{dist}(x, x^*)$ only in the analysis. Suppose a proposed flip reaches an unsatisfying assignment y outside the ball of radius $2H$ around x_0 . The triangle inequality gives

$$d(y) \geq \text{dist}(y, x_0) - \text{dist}(x_0, x^*) \geq 2H + 1 - h \geq h + 1.$$

Resetting to x_0 therefore strictly decreases this potential. Consequently, after the flip and any reset, each of the at least r favorable symbols still either terminates the walk or decreases d by at least one; every other symbol increases d by at most one. Every live assignment kept by the algorithm lies within distance $2H$ of x_0 , so its potential is at most $L = h + 2H$.

Let $a = \lambda/2$ and $b = 1/(2\lambda)$ as in Lemma B.1. On $\{1, \dots, L\}$ define the reflected comparison matrix

$$(Rf)(d) = af(d-1) + bf(d+1) \quad (d < L), \quad (Rf)(L) = af(L-1) + bf(L),$$

where $f(0) = 0$. Its powers preserve nonnegative increasing vectors. Exactly as in the barrier lemma, induction on the remaining horizon therefore bounds the powered mass of surviving length- T execution words by $e_h^\top R^T \mathbf{1}$. This comparison is valid after every history, including histories containing resets; the clause selector may retain the entire history. The spectral estimate in that lemma gives

$$e_h^\top R^T \mathbf{1} \leq \frac{\lambda^h}{\sqrt{\lambda^2 - 1}} \exp\left(-\frac{\pi^2 T}{2(2L+1)^2}\right).$$

Apply the source constraint to whole length- T execution words and sum. Since $L = h + 2H \leq 3H$, the resulting bound is at most $CD\lambda^h e^{-cT/H^2}$, proving (11) for $s = s_*$; the inclusion $\mathcal{Q}_{s,D} \subseteq \mathcal{Q}_{s_*,D}$ gives the claim for $s \geq s_*$.

Since $L \leq 3H$, summing the capped exponential tail yields $O_{k,r}(H^2(h + \log D + 1))$. More explicitly, for $A > 0$ and $q > 0$, $\sum_{T \geq 0} \min\{1, Ae^{-qT}\}$ is at most $1 + q^{-1}(1 + \max\{0, \log A\})$ by comparison with the integral of the decreasing summand. Here $q^{-1} = O(H^2)$ and $\log A = h \log \lambda + \log D - \frac{1}{2} \log(\lambda^2 - 1)$.

For the implementation claim, keep a list of coordinates on which the current assignment differs from x_0 , together with an inverse index for constant-time insertion and deletion. A flip updates this list in constant time. At a reset, restore and remove every listed coordinate. Each restored coordinate was flipped at least once in that attempt, so the reset uses no more variable updates than the attempt used source-driven flips. Summing over disjoint attempts proves the factor-two bound. The list has size at most $2H + 1$ before a reset. $\square$

C Proofs for restart policies

C.1 Necessity in Theorem 4.1

The proof in the main text establishes (i) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (iii) is immediate. Suppose now that $\alpha_s(x) = 1$. Exact prefix capacity gives a probability law $\mu \in \mathcal{Q}_{s,1}$ supported on $\mathcal{N}_x$. Fix an arbitrary deterministic restart policy. At each restart, begin an independent stream with law μ . This defines an adversarial source for the entire execution, not an assumption about the algorithm's access to independent randomness.

For any observed word w , the policy determines a unique segmentation into completed attempt words $b_1, \dots, b_j$ and an unfinished attempt word v . The resulting source therefore satisfies

$$\nu[w] = \mu[b_1] \cdots \mu[b_j] \mu[v] \leq P[b_1]^s \cdots P[b_j]^s P[v]^s = P[w]^s.$$

The formula is consistent on the prefix tree, including at restart boundaries. Every attempt follows an original nonterminating execution until the policy abandons it. Every prefix on which an attempt terminates has probability zero. The countable union of these prefixes still has probability zero, proving the converse and the stronger final assertion.

For completeness, the time bound in the sufficiency proof uses the horizon T , rather than the longest failed-attempt word: a successful path can avoid $\mathcal{B}$ and run longer than every word in that cover. If the longest cover word has length L , the slightly sharper bound $\Pr_\nu\{\tau_{\text{restart}} > T + kL\} \leq Dc^k$ also holds.

C.2 Proof of Theorem 4.2

The exact finite-horizon recursion is $V_0 = \mathbf{1}$ and $V_{t+1} = \min\{\mathbf{1}, RV_t\}$, with the minimum taken coordinatewise. Its decreasing limit V is a fixed point. Every feasible vector v in (12) satisfies $v \leq V_t$ by induction, hence $v \leq V$. Conversely, V is feasible. It is thus the componentwise greatest feasible vector and the unique maximizer of the sum objective.

Suppose $V(x) < 1$, and set

$$U = \{y : V(y) < 1\}, \quad S = X \setminus U, \quad A = R_{U,U}, \quad b = R_{U,S}\mathbf{1}.$$

We first show $\rho(A) < 1$. Otherwise take a nonnegative Perron eigenvector of A , normalize its maximum to one, and extend it by zero outside U . The extended vector w satisfies $0 \leq w \leq \mathbf{1}$ and $w \leq Rw$. It is feasible in (12) and has a coordinate equal to one inside U , contradicting $w \leq V$.

Since $V < 1$ on U and $V = 1$ on S , the fixed-point equations give

$$V_U = AV_U + b = (I - A)^{-1}b.$$

Write $G = (I - A)^{-1}$ and $c = e_x^\top Gb = V(x) < 1$. Under immediate restart on entry to S , the powered live matrix is $B = A + be_x^\top$. The rank-one inverse identity yields

$$(I - B)^{-1} = G + \frac{Gbe_x^\top G}{1 - c} \geq 0.$$

Thus $h = (I - B)^{-1}\mathbf{1} \geq \mathbf{1}$ and $Bh = h - \mathbf{1}$. For $H = \max_y h(y) > 1$, this gives $Bh \leq (1 - H^{-1})h$, so $\rho(B) < 1$. If $H = 1$, then $B\mathbf{1} = 0$ and termination occurs after one source-consuming step.

Summing powered probabilities over all surviving execution words gives

$$\Pr_\nu\{\tau_{\text{restart}} > t\} \leq \min\{1, De_x^\top B^t \mathbf{1}\}, \quad (25)$$

$$\mathbb{E}_\nu \tau_{\text{restart}} \leq Dh(x) = D \frac{e_x^\top (I - A)^{-1} \mathbf{1}}{1 - V(x)}. \quad (26)$$

For $H > 1$, the first bound is at most $Dh(x)(1 - H^{-1})^t$. Equation (26) follows by summing over $t \geq 0$ and using the inverse identity. These bounds use the global constraint only on whole execution words.

If the entries of R are rational, the LP, its optimizer, and the restart set can be computed in polynomial bit complexity in the explicit state-space input. This assertion requires exact representations of the powered coefficients; numerical approximations to arbitrary real values $\pi(a)^s$ do not supply such an input.

C.3 Proof of Proposition 4.3

Use source alphabet $\{0, 1, 2, 3\}$, initial state x , and the transition table below, where $\dagger$ denotes termination.

	0	1	2	3
x	$\dagger$	y	z	z
y	$\dagger$	y	y	y
z	z	$\dagger$	$\dagger$	$\dagger$

Under the uniform product law,

$$\mathbb{E}_P \tau = 1 + \frac{1}{4} \cdot 4 + \frac{1}{2} \cdot \frac{4}{3} = \frac{8}{3}.$$

At exponent $1/2$, each symbol has weight $1/2$. The original powered live matrix, in state order (x, y, z) , is

$$R = \begin{pmatrix} 0 & 1/2 & 1 \\ 0 & 3/2 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}.$$

For every $T \geq 1$, the failed length- T words F_T have total powered mass

$$Z_T^{\text{live}} := \sum_{w \in F_T} P[w]^{1/2} = \frac{1}{2}(3/2)^{T-1} + (1/2)^{T-1} \geq 1.$$

For $T = 1, 2$, the values are $3/2, 5/4$; for $T \geq 3$, the first term alone exceeds one. The original nontermination capacities are $(1/2, 1, 0)$.

For a finite cutoff T , put the probability law $q_T(w) = P[w]^{1/2}/Z_T^{\text{live}}$ on F_T . At a nonempty partial word u of length $j \leq T$, the ratio $q_T[u]/P[u]^{1/2}$ is zero or one of

$$\frac{(3/2)^{T-j}}{Z_T^{\text{live}}} \leq 2 \quad \text{on the } y \text{ branch}, \quad \frac{(1/2)^{T-j}}{Z_T^{\text{live}}} \leq 1 \quad \text{on the } z \text{ branch}.$$

At a complete block, this ratio is $1/Z_T^{\text{live}} \leq 1$. For any predetermined finite-cutoff sequence, concatenate independent blocks with these laws. Each complete block contributes a ratio at most one, and the final partial block contributes at most two. The resulting law $\mu \in \mathcal{Q}_{1/2,2}$ is supported on nontermination of the scheduled procedure. If the schedule has a final infinite cutoff, choose its first symbol to be 1, then choose independent uniform symbols from $\{1, 2, 3\}$ forever. The partial-prefix ratios in this final attempt are also at most two. Hence $\mu/2$ is a subprobability measure satisfying the unit prefix caps. Prefix-flow completion extends it to a probability law in $\mathcal{Q}_{1/2,1}$ while retaining nontermination mass at least $1/2$. The adversarial source may depend on the predetermined schedule.

The adaptive rule restarts immediately on entry to y . Its powered live matrix on (x, z) is $B = \begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix}$. The exact capped recursion gives $v_T(z) = 2^{-T}$ and $v_0(x) = v_1(x) = v_2(x) = 1$. For $T \geq 3$,

$$v_T(x) = \frac{1}{2}v_{T-1}(x) + 2^{-(T-1)} = T2^{1-T}.$$

The formula also holds at $T = 2$. Exact prefix capacity now gives $\max_{\nu \in \mathcal{Q}_{1/2,D}} \Pr_{\nu}\{\tau_{\text{adaptive}} > T\} = \min\{1, DT2^{1-T}\}$.

D Proofs for scheduling

D.1 Controlled prefix capacities

For $0 < s \leq 1$, take a finite state space, a common finite symbol alphabet with full-support probabilities $\pi(a)$, and a finite nonempty set F_x of available updates at each live state. Write $P = \pi^{\mathbb{N}}$ and let $\Phi_f(x, a)$ be the state obtained when update f uses symbol a at state x . Set $V_0(x) = 1$ at live states and $V_t(x) = 0$ at terminated states. At live states define

$$V_{t+1}(x) = \min \left\{ 1, \min_{f \in F_x} \sum_a \pi(a)^s V_t(\Phi_f(x, a)) \right\}. \quad (27)$$

Proposition D.1 (Optimal deterministic selection). *For every $D \geq 1$ and finite horizon T ,*

$$\min_{\Lambda} \max_{\nu \in \mathcal{Q}_{s,D}} \mathbb{P}_{\nu}^{\Lambda} \{\tau > T\} = \min\{1, DV_T(x_0)\}, \quad (28)$$

where Λ ranges over deterministic nonanticipating selectors. One selector depending only on current state and remaining time is optimal simultaneously for every $D \geq 1$ and every initial state.

This recursion evaluates a single globally constrained source; it does not impose independent conditional constraints after successive symbols.

Proof of Proposition D.1. Fix a finite horizon. At a live root, a prefix cover either cuts the root at cost one or covers the children after the chosen update. For a specified first update f , the selector can choose its continuation independently after each observed symbol a . The optimal continuation cost is $V_t(\Phi_f(x, a))$, multiplied by the edge cost $\pi(a)^s$. Minimizing over f and comparing with the root cut gives (27) by induction. At the horizon, the surviving event costs one at a live state and zero at a terminated state, exactly the stated boundary values. Each minimum is attained, and an attaining choice can depend only on current state and remaining time. These choices depend on s but not on D , and the same state-based rule is available from every initial state. Theorem 2.1 now turns the optimal cover cost into (28), since $c \mapsto \min\{1, Dc\}$ is nondecreasing.

This proof optimizes one globally constrained source. It does not reset the domination constant after a prefix or give the source independent conditional choices at each step. The selector has no additional private randomness; allowing trusted private coins would define a different optimization problem. $\square$

D.2 Optimal priority for heterogeneous retry tasks

Proof of Theorem 5.1. For a set S of unfinished tasks, the controlled recursion gives

$$V_{t+1}(S) = \min \left\{ 1, \min_{i \in S} ((W - w_i)V_t(S) + w_i V_t(S \setminus \{i\})) \right\}, \quad (29)$$

with $V_t(\emptyset) = 0$ and $V_0(S) = 1$ for nonempty S . Symbols with the same success status lead to the same state, so their powered masses add even when a selector remembers the individual labels. Proposition D.1 gives the exact optimum $\min\{1, DV_T(S)\}$.

We first record three monotonicities. Values decrease with t , because the recursion operator is monotone and $V_1 \leq V_0$. They increase when tasks are added. Indeed, suppose $R \subseteq S$ and argue by induction on t . An action $i \in R$ has at least its corresponding cost on R . An action $i \in S \setminus R$ has cost at least $WV_t(R) \geq V_t(R) \geq V_{t+1}(R)$, since $W \geq 1$. Taking the minimum and truncating at one preserve the comparison.

For fixed W , increasing a task's weight cannot increase the value. To see this, compare recursions in which its weight is a and $b \geq a$. By induction, the present-state values satisfy $X_a \geq X_b$, and the common value after completing that task is $Y \leq X_b$. The corresponding action costs differ by

$$(W - a)X_a + aY - ((W - b)X_b + bY) = (W - a)(X_a - X_b) + (b - a)(X_b - Y) \geq 0.$$

Other action costs are ordered by induction. These arguments apply to the abstract recursion for all $0 < w_i \leq W$ and $W \geq 1$, without requiring a modification of the reference source.

We prove a stronger invariant by induction on t : for distinct $i, j \in S$ with $a = w_i < b = w_j$,

$$(b - a)V_t(S) \leq bV_t(S \setminus \{j\}) - aV_t(S \setminus \{i\}). \quad (30)$$

It holds at $t = 0$. The inequality says exactly that the action for i has no larger cost than the action for j in (29). Equal-weight tasks are interchangeable. Thus, if the invariant holds at time t , a minimum-weight task is an optimal first action for horizon $t + 1$.

We shall use the following elementary truncation fact. If $0 \leq z \leq y \leq x$ and $(b - a)x \leq by - az$, then

$$(b - a) \min\{1, x\} \leq b \min\{1, y\} - a \min\{1, z\}. \quad (31)$$

If $x \leq 1$, nothing changes. If $y \geq 1$, the right side is at least $b - a$. Otherwise $x > 1 > y$, and the unchanged right side is at least $(b - a)x \geq b - a$.

Fix i, j as above, and choose a minimum-weight task $k \in S$. First suppose $k \notin \{i, j\}$ and put $c = w_k$. The optimal first action in S , $S \setminus \{i\}$, and $S \setminus \{j\}$ can be k . Before truncation, each value is the same nonnegative linear combination, with coefficients $W - c, c$, of the corresponding values with and without k at time t . Combine (30) for S and $S \setminus \{k\}$ to obtain the desired inequality before truncation. The three quantities have order $x \geq y \geq z$: adding a task increases the value, and replacing a weight a by b decreases it. Equation (31) proves the invariant after truncation.

The remaining case is $k = i$, since $a < b$ excludes $k = j$. Write

$$X = V_t(S), \quad Y = V_t(S \setminus \{j\}), \quad Z = V_t(S \setminus \{i\}), \quad H = V_t(S \setminus \{i, j\}).$$

The monotonicities give $X \geq Y \geq Z \geq H$. Set

$$x = (W - a)X + aZ, \quad y = (W - a)Y + aH, \quad z = (W - b)Z + bH.$$

The first two quantities are the optimal costs before truncation for S and $S \setminus \{j\}$. The third is a permissible cost for $S \setminus \{i\}$, obtained by choosing j . Hence its truncated value is an upper bound on $V_{t+1}(S \setminus \{i\})$. We have $x \geq y$ and

$$y - z = (W - a)(Y - Z) + (b - a)(Z - H) \geq 0.$$

Furthermore,

$$by - az - (b - a)x = (W - a)(bY - aZ - (b - a)X) \geq 0.$$

Apply (31), then replace its last truncated quantity by the no-larger actual value. This proves (30) at $t + 1$ and completes the induction. Thus increasing-weight priority is optimal at every horizon. Following that order leaves only suffixes of the sorted list and gives the stated $O(nT)$ recursion. Monotonicity of $x \mapsto \min\{1, Dx\}$ proves simultaneous optimality for every $D \geq 1$.

Finally, fix $q > 0$, let $W_q = \sum_a \pi(a)^q$, and consider the tilted law $\pi_q(a) = \pi(a)^q / W_q$. Under its independent stream, preassign independent streams to the tasks. The number of attempts for task i is geometric with success probability $\sum_{a \in B_i} \pi_q(a)$. The workloads are independent and finite almost surely. Every permitted selector uses exactly their sum: before all tasks complete it must attempt some unfinished task, and none can receive more than its preassigned workload. The next unrevealed symbol of any selected stream has law π_q , even when the selection uses all previous observations. The stopping-time distribution is therefore selector independent. Multiplication by W_q^T proves equality of surviving-prefix power sums. The identity $Z_T(q) = W_q \tilde{Z}_{T-1}(q)$ proves the same claim for the completed-step convention. $\square$

D.3 Equal stopping laws and different robustness

Proof of Theorem 5.2. Put

$$u = \sqrt{\pi(0)} = \frac{\sqrt{15}}{4}, \quad v = \sqrt{\pi(1)} = \frac{1}{4}.$$

We first construct ν_n . Through time $8n$, generate actions with probabilities $3/4$ for 0 and $1/4$ for 1 until the n th zero appears; after that zero, output only zeros through time $8n$. After time $8n$, generate independent actions according to P . This specifies one law on tapes, without reference to the selector.

A prefix of length at most $8n$ preceding the n th zero, with k zeros, satisfies

$$\frac{\nu_n[w]}{P[w]^{1/2}} = \left(\frac{3}{\sqrt{15}}\right)^k \leq 1.$$

Suppose the n th zero appears at time $L \leq 8n$. For a positive-mass prefix of length $t \in [L, 8n]$, the corresponding ratio is

$$\left(\frac{3}{\sqrt{15}}\right)^n u^{-(t-L)} \leq \left(\frac{3/4}{u^8}\right)^n < 1,$$

since $L \geq n$ and $u^8 = (15/16)^4 > 3/4$. Zero-mass prefixes satisfy the required bound automatically. After time $8n$, each new reference action multiplies the ratio by $\pi(a)^{1/2} \leq 1$. Thus every finite prefix satisfies the defining inequality for $\mathcal{Q}$. Under Λ_A , either fewer than n zeros have appeared by time $8n$, so an A task remains unfinished, or the B task is attempted only after the n th zero and subsequently sees zeros alone. In both cases $\tau > 8n$, proving (13).

For (14), consider an arbitrary $\nu \in \mathcal{Q}$ and use Λ_B . If no 1 appears during the first $4n$ actions, the tape begins with 0^{4n} , an event of ν -probability at most

$$u^{4n} = \left(\frac{225}{256}\right)^n.$$

Otherwise let $j \leq 4n$ be the first position containing 1. This completes B , and survival after $8n$ repairs requires fewer than n zeros among the remaining $\ell = 8n - j \geq 4n$ actions. Set $r = v/(3u) = 1/(3\sqrt{15})$. Their total power weight is at most

$$\begin{aligned} \sum_{k=0}^{n-1} \binom{\ell}{k} u^k v^{\ell-k} &\leq r^{-(n-1)} (v + ur)^\ell \\ &\leq r^{-(n-1)} (1/3)^{4n} = r \left(\frac{\sqrt{15}}{27}\right)^n. \end{aligned}$$

The first-1 prefix $0^{j-1}1$ has power weight $u^{j-1}v$. Applying the cylinder bound and summing over $j \leq 4n$ gives the claimed second term because

$$\sum_{j=1}^{4n} u^{j-1}v \leq \frac{v}{1-u} = 4 + \sqrt{15}.$$

This argument holds for every source in $\mathcal{Q}$, including the source ν_n used above. To obtain the simpler constant in (14), put $\alpha = \sqrt{15}/27$, $\beta = 225/256$, and $c = (4 + \sqrt{15})/(3\sqrt{15})$. The bound is $\beta^n + c\alpha^n$. Since $\alpha < \beta$ and $c\alpha/\beta = 256(4 + \sqrt{15})/18225 < 1/8$, it is at most $(9/8)\beta^n < 1$ for every $n \geq 1$.

For the lower bound in (14), take an i.i.d. source with probabilities u and $1 - u$. Both are at most the corresponding square-root reference probabilities, since $1 - u \leq v$. It belongs to $\mathcal{Q}$, and its prefix 0^{8n} leaves B unfinished under every selector. Its probability $u^{8n} = (15/16)^{4n}$ is therefore a lower bound for every selector, including the minimax optimum.

The selector invariance of the stopping law and all prefix power sums follows from Theorem 5.1, applied to these success sets. $\square$

E Proof of the composition theorem

The following exact formula includes Corollary 6.1. It also allows components of different sizes and arbitrary initial assignments. The update rule and the reference probabilities are those of Section 6.

Theorem E.1 (Exact robustness under arbitrary interleaving). *Fix $N \geq 3$, put $s = \Delta_N$, $\theta = \pi/N$, and $r = 7^s$, and define*

$$V_N(k) = \begin{cases} 0, & k = 0, \\ r^{k-N+1} \frac{\sin(k\theta)}{\sin \theta}, & 1 \leq k < N-1, \\ 1, & k \geq N-1. \end{cases}$$

Let the graph be a disjoint union of finitely many connected components, each with at least N vertices. Let K_0 count the ones in the initially nonconstant components. For every deterministic nonanticipating edge selector and every $D \geq 1$,

$$\max_{\nu \in \mathcal{Q}_{s,D}} \nu\{\tau = \infty\} = \min\{1, DV_N(K_0)\}. \quad (32)$$

One source depending only on N, K_0, D attains this value simultaneously for every allowed graph, initial assignment with count K_0 , and selector. Under the independent reference source,

$$\mathbb{E}_P \tau \leq \frac{25}{12} K_0. \quad (33)$$

Proof. Write $V = V_N$ and let

$$d = (49/100)^s, \quad b = 2(1/4)^s, \quad u = (1/100)^s, \quad a = \sqrt{du} = (7/100)^s.$$

The equation defining s is $b + 2a \cos \theta = 1$. Its left side is strictly decreasing in s , exceeds one at $s = 1/2$, and is below one at $s = 1$. Thus $1/2 < s < 1$. In particular, $b < 1$, $d = ar$, $u = a/r$, and $r > \sqrt{7} > 2 \cos \theta$.

Set $h(k) = r^k \sin(k\theta)$ for $0 \leq k \leq N$. The sine addition formula gives

$$dh(k-1) + bh(k) + uh(k+1) = h(k), \quad 1 \leq k < N.$$

Moreover, h is strictly increasing on $1, \dots, N-1$: for $1 \leq k \leq N-2$, $\sin(k\theta) \leq 2 \cos \theta \sin((k+1)\theta)$ and $r > 2 \cos \theta$. Hence $0 < V(k) < 1$ for $1 \leq k < N-1$, and the same harmonic identity holds for V there. At the upper boundary it gives

$$dV(N-2) + b = 1. \quad (34)$$

At any history, let K count the ones in components that are still nonconstant. A downward action sends K to $K-1$; either neutral action leaves it unchanged. An upward action sends K to $K+1$, unless it makes an L -vertex component all ones, in which case K becomes $K+1-L$. Such a completion requires $K \geq L-1 \geq N-1$. Thus below $N-1$ the count has the same increments for every selector, even though this need not hold above that level.

For the upper bound, only $1 \leq K_0 < N-1$ requires proof. Cover each nonterminating tape at its first visit to $K = N-1$, or at depth T if it has not yet visited that level. Before the first visit, the powered count matrix on $1, \dots, N-2$ is tridiagonal with lower diagonal d , diagonal b , and upper diagonal u . Call it A . Diagonal similarity to a symmetric tridiagonal matrix gives

$$\rho(A) = b + 2a \cos \frac{\pi}{N-1} < 1.$$

The harmonic identity with boundary values $V(0) = 0$ and $V(N - 1) = 1$ therefore makes the total powered mass of first visits to $N - 1$ equal to $V(K_0)$. The remaining live prefixes at depth T have powered mass $e_{K_0}^T A^T \mathbf{1}$, which tends to zero. Applying the prefix bound to this finite cover and letting $T \rightarrow \infty$ gives $\nu\{\tau = \infty\} \leq \min\{1, DV(K_0)\}$.

For the matching source, suppose $K_0 > 0$. An auxiliary law q maintains a counter $Y_0 = K_0$. Let $\delta(z) \in \{-1, 0, 1\}$ be the increment assigned to action label z . When $1 \leq Y_t < N - 1$, use

$$q(z \mid Y_t) = \pi(z)^s \frac{V(Y_t + \delta(z))}{V(Y_t)}.$$

When $Y_t \geq N - 1$, give each neutral label probability $(1/4)^s$, the downward label probability $1 - b$, and the upward label probability zero. Update $Y_{t+1} = Y_t + \delta(z)$. The harmonic identity normalizes these probabilities below $N - 1$, and normalization above that level is immediate. The downward probability at one is zero, so Y_t stays positive.

Every upward move starts below $N - 1$, so it cannot finish a component. A downward completion produces an all-zero component and has no additional effect on the count. Consequently $K_t = Y_t$ for every graph, initial assignment, and selector under consideration. Every tape in the support of q is nonterminating for all these processes.

Equation (34), together with $V(N - 2) < 1$, gives $1 - b = dV(N - 2) < d$. It follows in the upper range as well as the lower range that

$$q(z \mid Y_t) \leq \pi(z)^s \frac{V(Y_{t+1})}{V(Y_t)}.$$

Multiplying along a prefix yields

$$q[w] \leq P[w]^s \frac{V(Y_{|w|})}{V(K_0)} \leq \frac{P[w]^s}{V(K_0)}. \quad (35)$$

Put $p = \min\{1, DV(K_0)\}$. The subprobability law $\eta = pq$ has mass p and obeys $\eta[w] \leq DP[w]^s$. It extends to a probability law ν obeying the same bound: at each node, distribute its assigned mass among the children, preserving their η masses and respecting their capacities $DP[wz]^s$. This is possible because the lower bounds sum to $\eta[w]$, whereas the capacities sum to $DP[w]^s \sum_z \pi(z)^s \geq DP[w]^s$. Starting with root mass one and using a fixed label order gives an extension depending only on N, K_0, D . Its common nontermination mass is at least p , proving equality in (32). If $K_0 = 0$, all components are already constant and the assertion is immediate. When $K_0 \geq N - 1$ and $D = 1$, $p = 1$ and q itself is the required source.

Finally, under P , a live update has downward drift at least $49/100 - 1/100 = 12/25$ in K ; finishing an all-one component only increases that downward drift. Summing the conditional drift bound up to $\tau \wedge t$ gives $(12/25)\mathbb{E}_P(\tau \wedge t) \leq K_0$. Letting $t \rightarrow \infty$ proves (33). $\square$

For one component starting with one vertex equal to one, $V_N(1) = 7^{-\Delta_N(N-2)}$. For $N - 1$ such components, $V_N(N - 1) = 1$ and the same source forces nontermination under every interleaving. Equation (33) gives the reference mean bound. This proves Corollary 6.1.

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AI methodologies

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